

Experiment 2: Basic Image Handling and Processing Operations

Aim

To read an image using Python and apply Gaussian Blur to the image.

Algorithm

1. Import the OpenCV (cv2) library.
2. Read the input image using cv2.imread().
3. Apply Gaussian Blur using cv2.GaussianBlur() with a kernel size of (5,5).
4. Display the original image.
5. Display the blurred image.
6. Wait for a key press and close all display windows.

Python Code

```
1  import sys
2  from pathlib import Path
3  import cv2
4
5
6  def blur_image(input_path: str, show_windows: bool = False) -> None:
7      image = cv2.imread(input_path)
8
9      if image is None:
10         raise FileNotFoundError(f"Could not read image: {input_path}")
11
12         blurred_image = cv2.GaussianBlur(image, (5, 5), 0)
13
14         input_file = Path(input_path)
15         output_path = input_file.with_name(f"{input_file.stem}_blurred{input_file.suffix}")
16         cv2.imwrite(str(output_path), blurred_image)
17
18         print(f"Input image read from: {input_path}")
19         print(f"Blurred image saved as: {output_path}")
20
21         if show_windows:
22             cv2.imshow("Original Image", image)
23             cv2.imshow("Blurred Image", blurred_image)
24             cv2.waitKey(0)
25             cv2.destroyAllWindows()
26
27
28  def main() -> None:
29      input_file = sys.argv[1] if len(sys.argv) > 1 else "input.jpg"
30      show_windows = "--show" in sys.argv[1:]
31      blur_image(str(Path(input_file)), show_windows=show_windows)
32
33
34  if __name__ == "__main__":
35      main()
36
```

INPUT



OUTPUT



Result

The input image was successfully read using Python, and Gaussian Blur was applied to the image successfully.